

Problem Set 7

A. Cops And Robbers

tianzheyu submissions							
#	When	Who	Problem	Lang	Verdict	Time	Memory
383693097	Jul/21/2026 20:50 ^{UTC+10}	tianzheyu	E - Cops And Roobers	C++23 (GCC 14-64, msys2)	Accepted	62 ms	24900 KB

<https://codeforces.com/submissions/tianzheyu#>

Process

Constructed a flow network with node capacities by splitting grid cells into in and out nodes to represent the barricade costs. Added a super sink for the grid boundaries and ran Dinic's algorithm to find the minimum cut.

Challenges and Overcoming

When coding the boundary check to connect edge cells to the super sink, I ran into a compilation bug stating: "expression must be a modifiable lvalue". I found it when the compiler highlighted my conditional statement: `if (i == 0 || j == 0 || i = m - 1 || j == n - 1)`. I had accidentally used `=` instead of `==` for the row boundary check. Because logical `||` has higher precedence than `=`, the compiler tried to assign the value `m - 1` to the boolean result of the earlier conditions, which is an rvalue. Next time to prevent this from happening, I will pay extra care for the condition statements.